

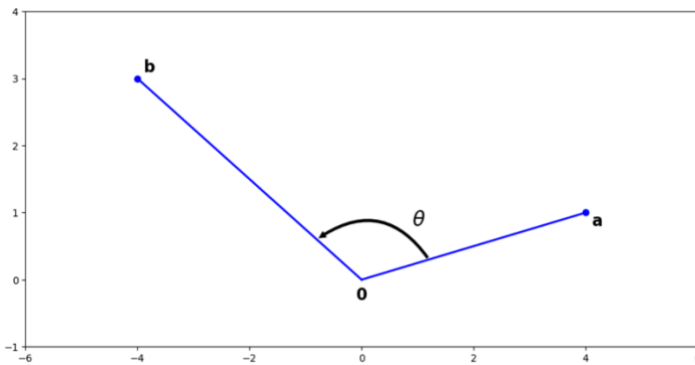
Lengths and Inner Products

Geometry of Vectors in $\mathbb{R}^n$

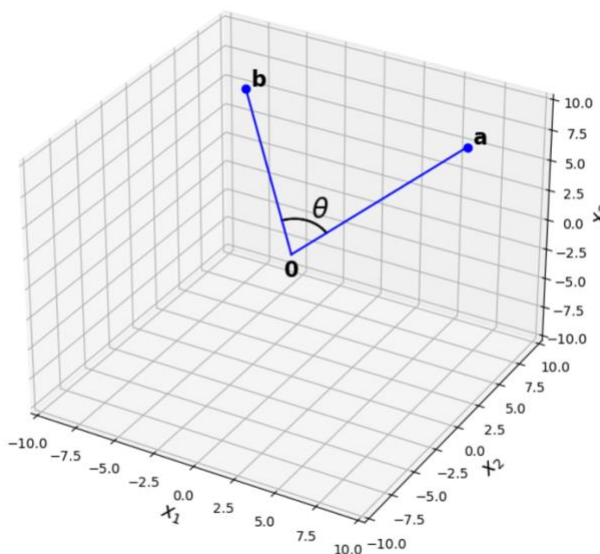
Let's consider some geometric questions involving vector lengths and angles.

In particular, we will take familiar notions and reformulate them in terms of vectors in arbitrary dimension.

Example: Let's start with a simple example in $\mathbb{R}^2$. How would you determine the angle θ ?



Let's go up a dimension: to $\mathbb{R}^3$. How would you determine the angle θ ?



This seems more challenging, but we could probably still do it.

If we go up one more dimension, to $\mathbb{R}^4$, then it's more challenging to solve this problem visually.

What we need are ways to capture simple notions in spaces of arbitrary dimension:

- length,
- distance,
- orthogonality (perpendicularity), and
- angle.

Interestingly, it turns out that these notions all depend on one key notion: the *inner product*.

Inner (Dot) Product

Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ be n -dimensional column vectors. In other words, they are matrices.

Remember: while you can add two vectors together to get another vector, you can **never multiply** two vectors together!

Definition:

Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ be n -dimensional column vectors. The inner (dot) product of $\mathbf{u}$ and $\mathbf{v}$ is the number $\mathbf{u}^T \mathbf{v}$ such that:

$$\mathbf{u}^T \mathbf{v} = [u_1 \quad u_2 \quad \dots \quad u_n] \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = u_1 v_1 + u_2 v_2 + \dots + u_n v_n = \sum_{i=1}^n u_i v_i$$

Inner product is denoted by $\mathbf{u} \cdot \mathbf{v}$

Compute the inner product of $\mathbf{u} = \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 2 \\ 4 \\ -3 \end{bmatrix}$

Example:

Assume we have three goods to buy and sell. Their prices are known per unit.

- Price vector: $\mathbf{p} = \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix}$
- Quantities vector: $\mathbf{q} = \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix}$

The quantities are positive when we sell and negative when we buy.

Then, the total income is the inner product $\mathbf{q} \cdot \mathbf{p}$

How would you interpret the signs of the inner product $\mathbf{q} \cdot \mathbf{p}$ in this context?

What does it mean when the inner product is 0?

Properties of inner products

Let u, v and w be vectors in $\mathbb{R}^n$ and let c be a scalar. Then

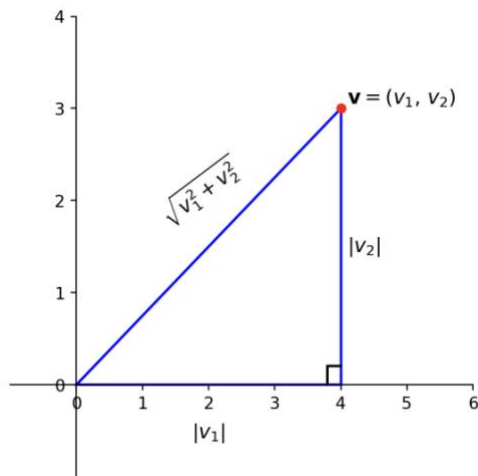
- $\mathbf{u}^T \mathbf{v} = \mathbf{v}^T \mathbf{u}$ (Symmetry)
- $(\mathbf{u} + \mathbf{v})^T \mathbf{w} = \mathbf{u}^T \mathbf{w} + \mathbf{v}^T \mathbf{w}$ (distributive property)
- $(c\mathbf{u})^T \mathbf{v} = c(\mathbf{u}^T \mathbf{v}) = \mathbf{u}^T (c\mathbf{v})$ (associative property)
- $\mathbf{u}^T \mathbf{u} \geq 0$ (non-negativity)

NOTE: $\mathbf{u}^T \mathbf{u} = 0$ if and only if $\mathbf{u} = \mathbf{0}$

Why is inner product symmetric?

Vector Length

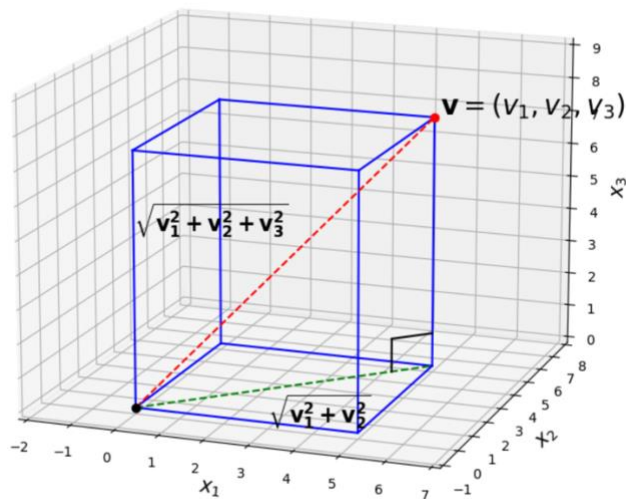
How do we measure the length of a vector? (think: how big is the vector?)



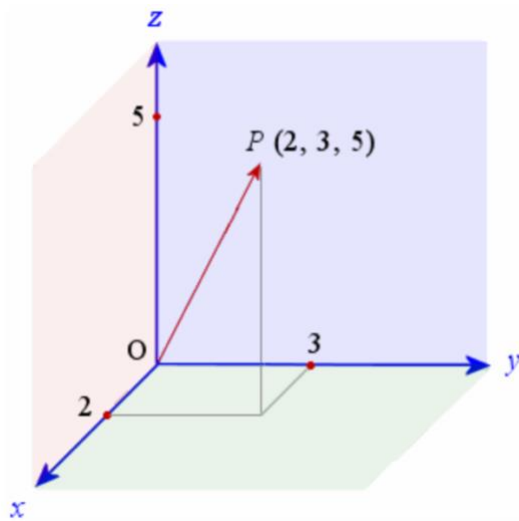
In $\mathbb{R}^2$ length is $\sqrt{v_1^2 + v_2^2}$

What about $\mathbb{R}^3$?

In $\mathbb{R}^3$ length is $\sqrt{v_1^2 + v_2^2 + v_3^2}$



Example: Let's try to find the length of vector $\mathbf{p} = \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix}$



Vector length (Norm)

The *length or norm or magnitude* of a vector $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$ is a non-negative scalar

$$\|\mathbf{v}\| \text{ defined by: } \|\mathbf{v}\| = \sqrt{\mathbf{v}^T \mathbf{v}} = \sqrt{\sum_{i=1}^n v_i^2} = \sqrt{v_1^2 + v_2^2 + \dots + v_n^2}$$

Which vector is longer?

$$\mathbf{a} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{bmatrix} \text{ or } \mathbf{b} = \begin{bmatrix} 3 \\ 3 \\ 3 \\ 3 \\ 3 \end{bmatrix}$$

Normalization to unit length

The word “unit” is always indicating that some measurement equals “one”.

Unit Vector

A unit vector $\mathbf{u}$ is a vector whose length equals 1.i.e. $\|\mathbf{u}\|=1$

For any scalar c , the length of $c\mathbf{v}$ is $|c|$ times the length of $\mathbf{v}$. That is,

$$\|c\mathbf{v}\| = |c|\|\mathbf{v}\|$$

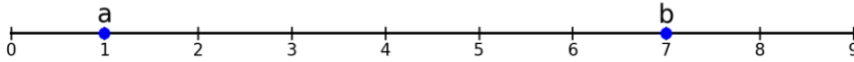
If we divide a non-zero vector by its length, we get a unit vector:

$$\mathbf{u} = \frac{1}{\|\mathbf{v}\|} \mathbf{v}$$

- If $\mathbf{u}$ is a unit vector, what is the length of $-2\mathbf{u}$?
- Let $\mathbf{v} = \begin{bmatrix} 4 \\ -2 \\ 2 \\ 1 \end{bmatrix}$, find the unit vector in the direction of $\mathbf{v}$.

Distance

In $\mathbb{R}^1$ the distance between two points a and b is $|a - b|$



The same can be extended to $\mathbb{R}^n$

For $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$ the distance between $\mathbf{u}$ and $\mathbf{v}$, written as $\mathbf{dist}(\mathbf{u}, \mathbf{v})$, is the length of the vector $\mathbf{u} - \mathbf{v}$. That is,

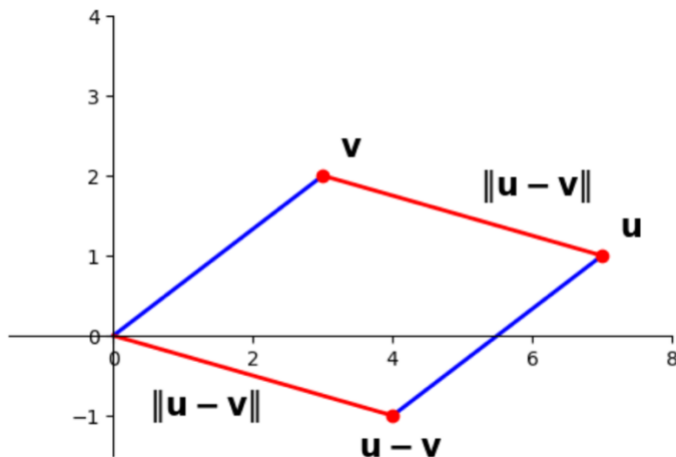
$$\mathbf{dist}(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|$$

Algebraically, it agrees with the usual formulas for the Euclidean distance between two points. The usual formula is:

$$\mathbf{dist}(\mathbf{u}, \mathbf{v}) = \sqrt{(v_1 - u_1)^2 + (v_2 - u_2)^2 + \cdots + (v_n - u_n)^2}$$

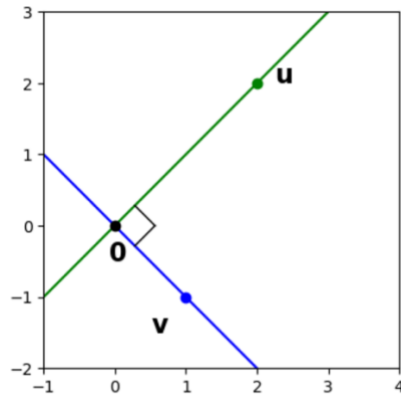
This is equal to:

$$\begin{aligned} \|\mathbf{u} - \mathbf{v}\| &= \sqrt{(\mathbf{u} - \mathbf{v})^T (\mathbf{u} - \mathbf{v})} \\ &= \sqrt{[\mathbf{u}_1 - \mathbf{v}_1 \quad \mathbf{u}_2 - \mathbf{v}_2 \quad \cdots \quad \mathbf{u}_n - \mathbf{v}_n] \begin{bmatrix} \mathbf{u}_1 - \mathbf{v}_1 \\ \mathbf{u}_2 - \mathbf{v}_2 \\ \vdots \\ \mathbf{u}_n - \mathbf{v}_n \end{bmatrix}} \end{aligned}$$

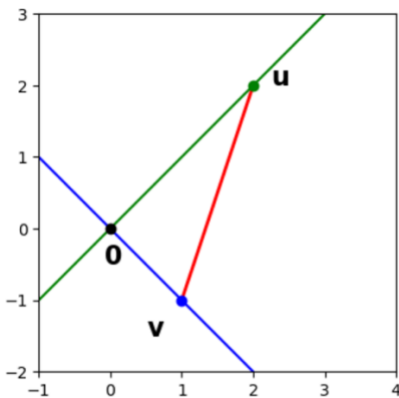


Orthogonality

Two vectors $\mathbf{u}$ and $\mathbf{v}$ are perpendicular or orthogonal if they form a right angle at the origin.

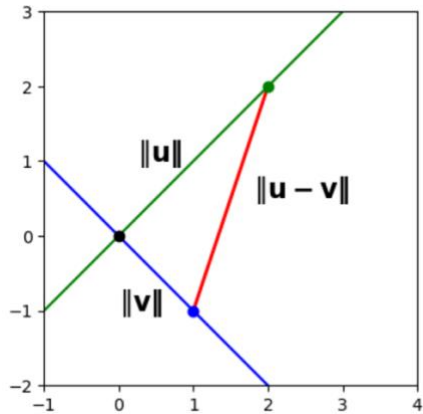


Draw the line connecting $\mathbf{u}$ and $\mathbf{v}$. Then $\mathbf{u}$ and $\mathbf{v}$ are perpendicular if and only if this is a right triangle, meaning that the Pythagorean Theorem is satisfied.



What is the length of the red side of the triangle?

According to the definitions we've developed today, it is $\|\mathbf{u} - \mathbf{v}\|$.



So the blue and green lines are perpendicular if and only if:

$$\|\mathbf{u} - \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$$

let's use the algebraic properties of the inner product to calculate the squared distance from $\mathbf{u}$ to $\mathbf{v}$ in another way:

$$\|\mathbf{u} - \mathbf{v}\|^2 = (\mathbf{u} - \mathbf{v})^T (\mathbf{u} - \mathbf{v})$$

$$\|\mathbf{u} - \mathbf{v}\|^2 = (\mathbf{u}^T - \mathbf{v}^T)(\mathbf{u} - \mathbf{v})$$

$$\|\mathbf{u} - \mathbf{v}\|^2 = \mathbf{u}^T (\mathbf{u} - \mathbf{v}) - \mathbf{v}^T (\mathbf{u} - \mathbf{v})$$

$$\|\mathbf{u} - \mathbf{v}\|^2 = \mathbf{u}^T \mathbf{u} - \mathbf{u}^T \mathbf{v} - \mathbf{v}^T \mathbf{u} + \mathbf{v}^T \mathbf{v}$$

$$\|\mathbf{u} - \mathbf{v}\|^2 = \|\mathbf{u}\|^2 - \|\mathbf{v}\|^2 - 2\mathbf{u}^T \mathbf{v}$$

****Remember, inner product is symmetric**

We now have two equations for the square of the distance between $\mathbf{u}$ and $\mathbf{v}$

$$\|\mathbf{u} - \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$$

$$\|\mathbf{u} - \mathbf{v}\|^2 = \|\mathbf{u}\|^2 - \|\mathbf{v}\|^2 - 2\mathbf{u}^T \mathbf{v}$$

Thus, if $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$, are perpendicular, then $\mathbf{u}^T \mathbf{v} = \mathbf{0}$

Definition: Two vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$ are orthogonal to each other if $\mathbf{u}^T \mathbf{v} = \mathbf{0}$

NOTE:

Inner product of two vectors is 0 when the angle between them is 90° . Then $\cos\theta = 0$.

If $\mathbf{u}^T \mathbf{v} \neq 0$ then:

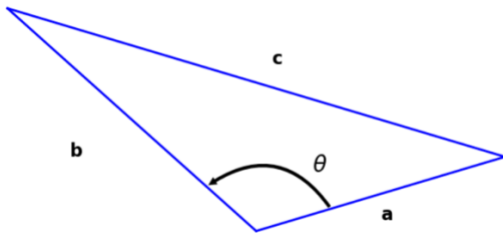
- If $\mathbf{u}^T \mathbf{v} > 0$, then the angle is less than 90°
- If $\mathbf{u}^T \mathbf{v} < 0$, then angle is greater than 90°

Angle between two vectors

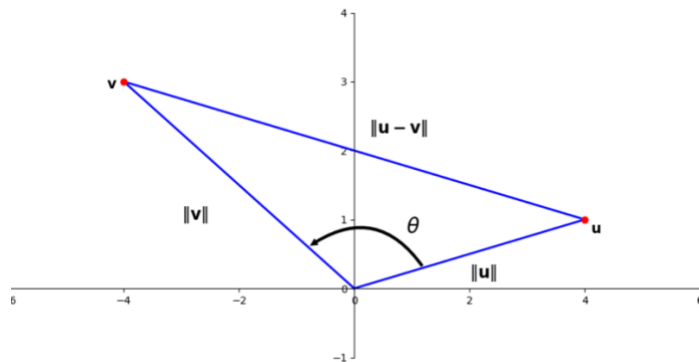
There is an important connection between the inner product of two vectors and the angle between them. This connection is very useful (e.g., in visualizing data mining operations).

Recall the **law of cosines**:

$$c^2 = a^2 + b^2 - 2ab \cos \theta$$



Let's interpret this in terms of vectors $\mathbf{u}$ and $\mathbf{v}$



Applying the law of cosines we get:

$$\|\mathbf{u} - \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\|\mathbf{u}\|\|\mathbf{v}\| \cos \theta$$

We already know that:

$$\|\mathbf{u} - \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\mathbf{u}^T \mathbf{v}$$

This gives us

$$\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\mathbf{u}^T \mathbf{v} = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\|\mathbf{u}\|\|\mathbf{v}\| \cos \theta$$

So,

$$\mathbf{u}^T \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta$$

This then follows:

$$\frac{\mathbf{u}^T \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|} = \cos \theta$$

Equivalently:

$$\left(\frac{\mathbf{u}^T}{\|\mathbf{u}\|} \right) \left(\frac{\mathbf{v}}{\|\mathbf{v}\|} \right) = \cos \theta$$

Note that $\frac{\mathbf{u}^T}{\|\mathbf{u}\|}$ and $\frac{\mathbf{v}}{\|\mathbf{v}\|}$ are both unit vectors.

So, we have the very simple rule, that for two unit vectors, their inner product is the cosine of the angle between them!

Example:

Let $\mathbf{u} = \begin{bmatrix} .8 \\ .6 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} .28 \\ .96 \end{bmatrix}$

They are both unit vectors.

So, $\mathbf{u}^T \mathbf{v} = (0.8 \cdot 0.28) + (0.6 \cdot 0.96) = 0.8$

So, $\cos \theta = 0.8$. We can now use this to find θ

Understanding the distances and angles between vectors allow us to evaluate the "similarity" between two vectors. This is the first step in two kinds of learning algorithms: -nearest neighbors and -means clustering.

Given, vector $\mathbf{v} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$. Can you find a vector orthogonal to it?

